Group No.:5

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Institute Rule Based FAQ Chatbot

Problem Statement

Educational institutions receive a large number of repetitive queries from students and visitors regarding admissions, course details, fee structure, institute timings, and campus facilities.

Handling these queries manually creates several problems:

- Administrative staff must repeatedly answer the same questions.
- Students may face delays in receiving information.
- Communication channels such as phone calls or emails become overloaded.
- Information may not always be available outside office hours.

Therefore, there is a need for an **automated system that can answer common queries instantly**.

The Institute FAQ Chatbot addresses this problem by providing an interactive platform where users can ask questions and receive immediate responses without requiring human intervention.

With the growing number of student inquiries in educational institutions, providing quick and accurate information has become essential. Chatbots are intelligent systems designed to simulate conversation with users and provide automated responses to queries.

This project focuses on developing an **Institute FAQ Chatbot** that helps students obtain information related to institute timings, admissions, courses, fees, placements, and other facilities. The chatbot provides instant responses using a keyword-based matching approach, making it a simple and effective solution for frequently asked questions.

Objectives

The main objectives of this project are:

- To develop a web-based chatbot for answering common institute-related queries.
- To reduce manual effort in handling repetitive questions.
- To provide instant information to students and visitors.
- To design a user-friendly chat interface.
- To demonstrate the use of JavaScript for simple natural language interaction.

Methodology

The development of the Institute FAQ Chatbot follows a simple and structured methodology.

Step 1: Interface Design

A chatbot interface is designed using **HTML and CSS**.
The interface contains:

- Chat header
- Chat message display area
- User input box
- Send button

Step 2: Keyword-Based Knowledge Base

A set of **keyword groups** is created in JavaScript.
Each group contains:

- Related keywords
- Corresponding chatbot response

Example:

keywords: ["fees", "tuition", "payment"]
response: "Fee structure varies by course."

Step 3: User Input Processing

When the user types a message:

1. The message is converted into lowercase.
2. The system scans the message for keywords.

Step 4: Response Matching

If a keyword is detected, the chatbot returns the associated response.

Example:

User Input:
"Tell me about fees"

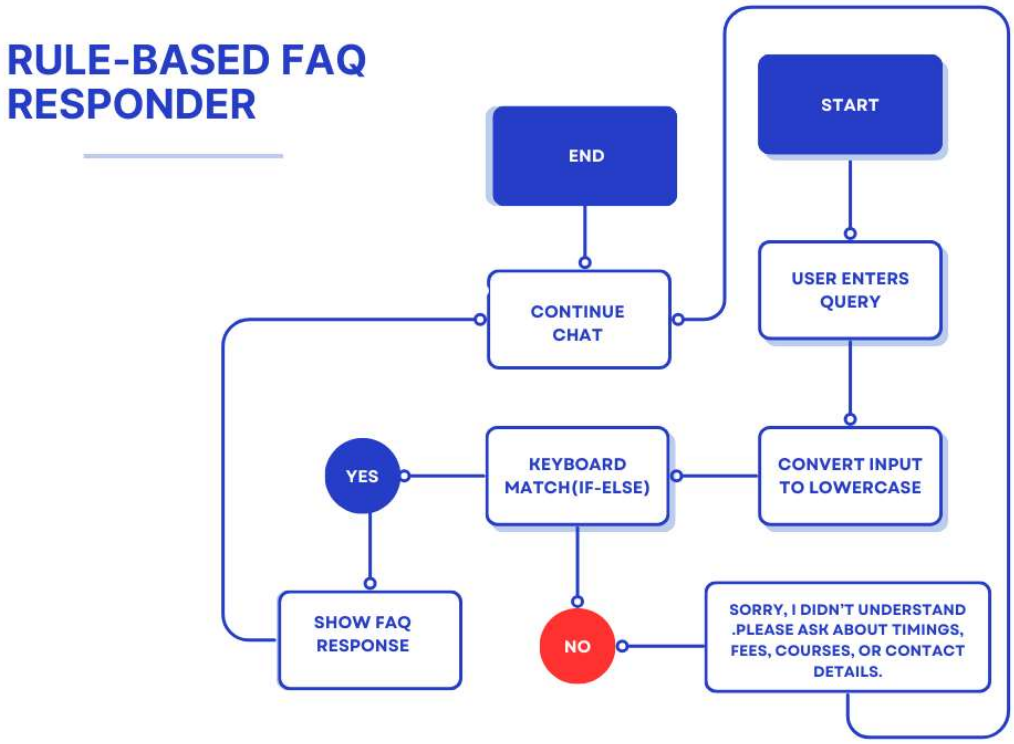
Detected Keyword:
fees

Chatbot Response:
"Fee structure varies by course."

Step 5: Default Response

If no keyword is matched, the chatbot responds with a default message asking the user to rephrase the question.

Flowchart:



Technologies Used:

1 HTML

HTML (HyperText Markup Language) is used to create the structure of the chatbot interface, including:

- Chat container
- Message display area
- Input field
- Send button

2 CSS

CSS is used to enhance the visual appearance of the chatbot interface. It includes:

- Gradient background
- Chat bubble styling
- Animations for messages
- Scrollbar customization
- Responsive layout

3 JavaScript

JavaScript is used to implement the chatbot logic. It performs tasks such as:

- Receiving user input
- Matching keywords
- Generating responses
- Displaying messages in the chat interface

System Design

1 User Interface

The chatbot interface consists of three main sections:

1. **Chat Header**
   - Displays the chatbot title: *Symbiosis Institute Chatbot*.
2. **Chat Box**
   - Shows conversation between the user and chatbot.
3. **Chat Input**
   - Text field for typing questions.
   - Send button to submit the message.

2 Chatbot Working Mechanism

The chatbot works using **keyword-based semantic matching**.

The system stores multiple categories of keywords and corresponding responses.

Example categories include:

- Timing
- Courses

- Admission
- Fees
- Scholarship
- Contact
- Location
- Hostel
- Placement
- Library
- Sports
- Cafeteria
- Transport

When a user enters a message:

1. The input text is converted to lowercase.
2. The program checks whether the text contains any predefined keywords.
3. If a keyword matches, the chatbot returns the corresponding response.
4. If no match is found, the chatbot returns a default response.

Functional Components

1 Keyword Groups

The chatbot stores questions in **keyword groups**.

Example:

- Keywords:
  fees, payment, cost, tuition
- Response:
  "Fee structure varies by course. Please check the official website or contact the administration office."

This allows the chatbot to understand different ways of asking the same question.

2 Message Handling

The chatbot performs two main actions:

1. Add User Message

- Displays the user's question in the chat interface.

2. Generate Bot Response

- Searches for matching keywords
- Displays the corresponding answer.

3 Default Response

If the chatbot cannot identify the query, it replies:

"Sorry, I didn't understand. Please ask about timings, fees, courses, admissions, placements, contact details, or other facilities."

This ensures the conversation continues smoothly.

Features of the Chatbot

- Interactive chat interface
- User-friendly design
- Fast response generation
- Keyword-based semantic matching
- Handles multiple categories of questions
- Automatic scrolling chat window
- Responsive and animated UI

Advantages

- Provides instant answers to frequently asked questions.
- Reduces workload on administrative staff.
- Improves accessibility of information for students.
- Easy to implement and deploy.
- Can run directly in a web browser.

Limitations

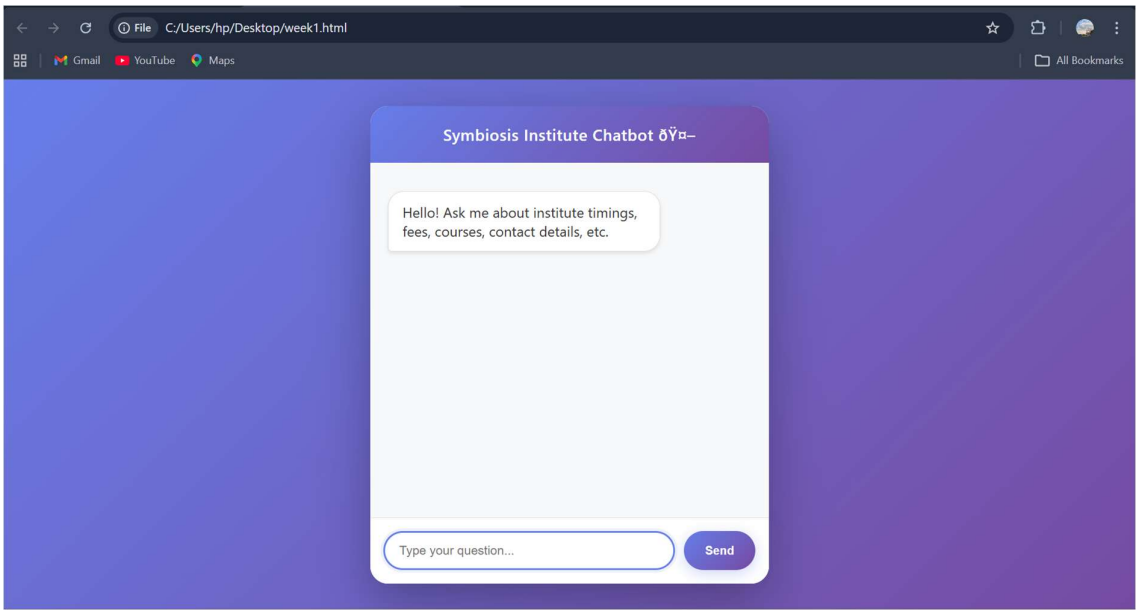
- The chatbot uses **keyword matching instead of AI-based NLP**.
- It may fail to understand complex or unusual questions.
- Responses are predefined and cannot generate dynamic answers.

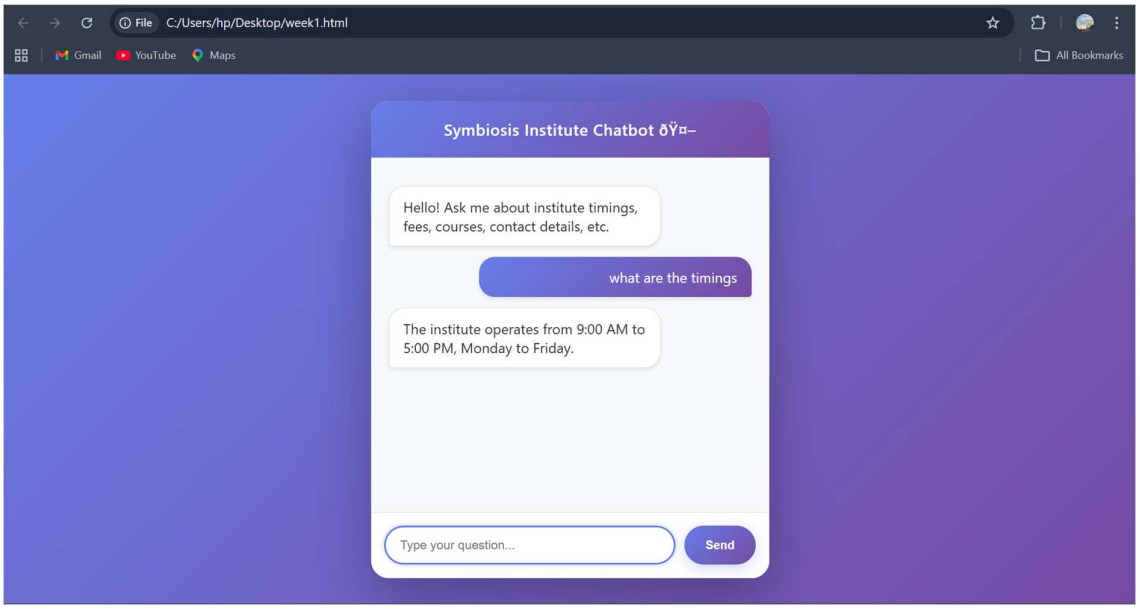
Future Enhancements

The chatbot can be improved by adding:

- **Natural Language Processing (NLP)** for better understanding of queries.
- **Machine Learning models** for intelligent responses.
- **Database integration** for dynamic information retrieval.
- **Voice-based interaction**.
- **Integration with institute websites and mobile apps.**

Output Screenshots:





Conclusion

The Institute FAQ Chatbot is a simple yet effective system that helps students quickly obtain information about the institute. By using HTML, CSS, and JavaScript, the chatbot provides an interactive platform for answering frequently asked questions. Although it currently uses keyword matching, it can be further enhanced with AI technologies to provide more intelligent and dynamic responses.

When a user enters a question, the system searches for specific keywords and returns predefined answers related to those keywords.

The objective of this system is to reduce the workload of administrative staff while providing instant information to users through a simple and interactive chat interface. The chatbot improves accessibility, efficiency, and user experience by enabling automated communication.